# Experiment 10:

## 1. Verification of Snort Installation (snort -V)

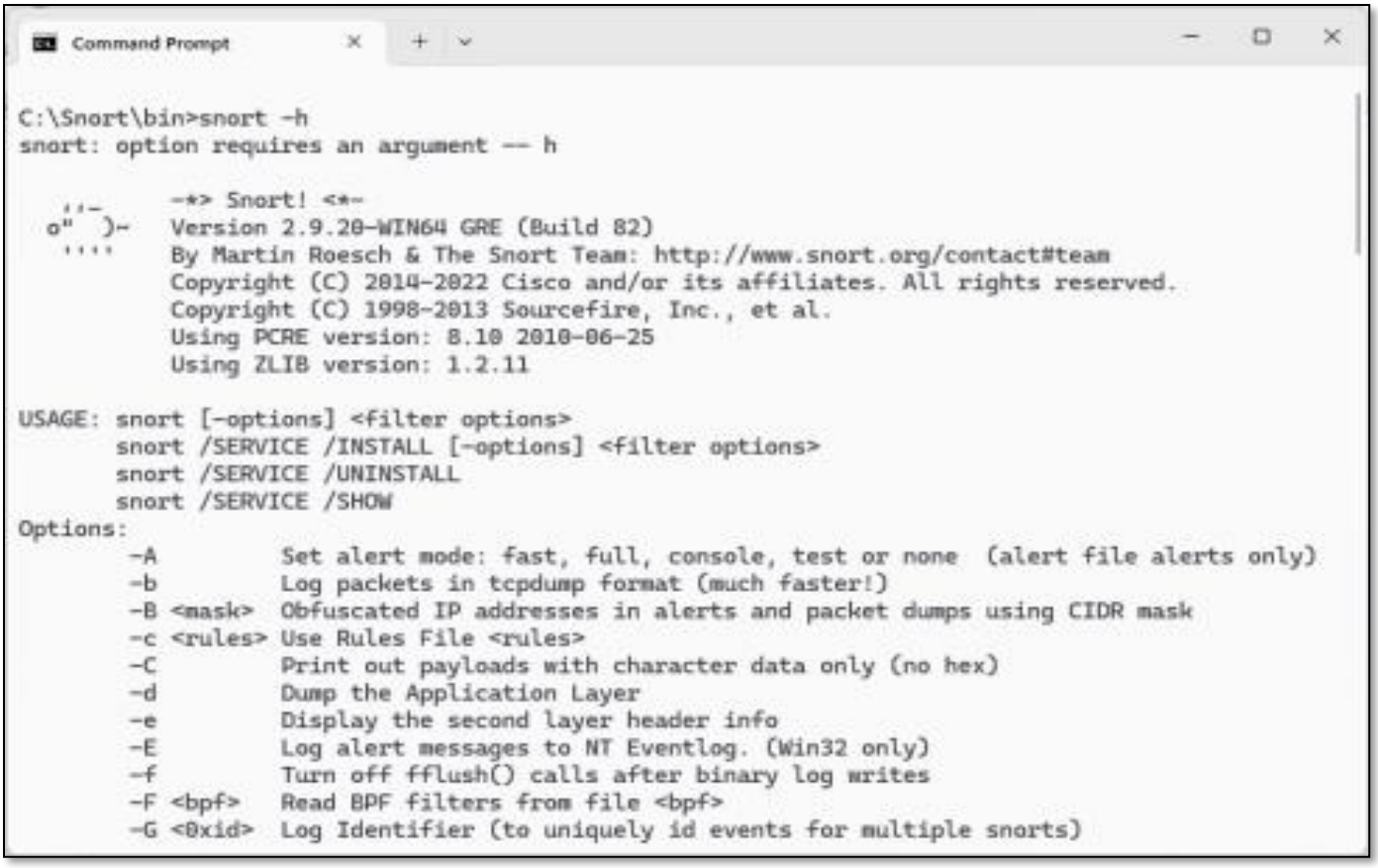



### Observation:

The command displays the installed Snort version, build information, copyright details, and supporting library versions. This confirms that Snort is installed and its version can be checked successfully.

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## 2. Snort Help Menu and Command-Line Options (snort -h)

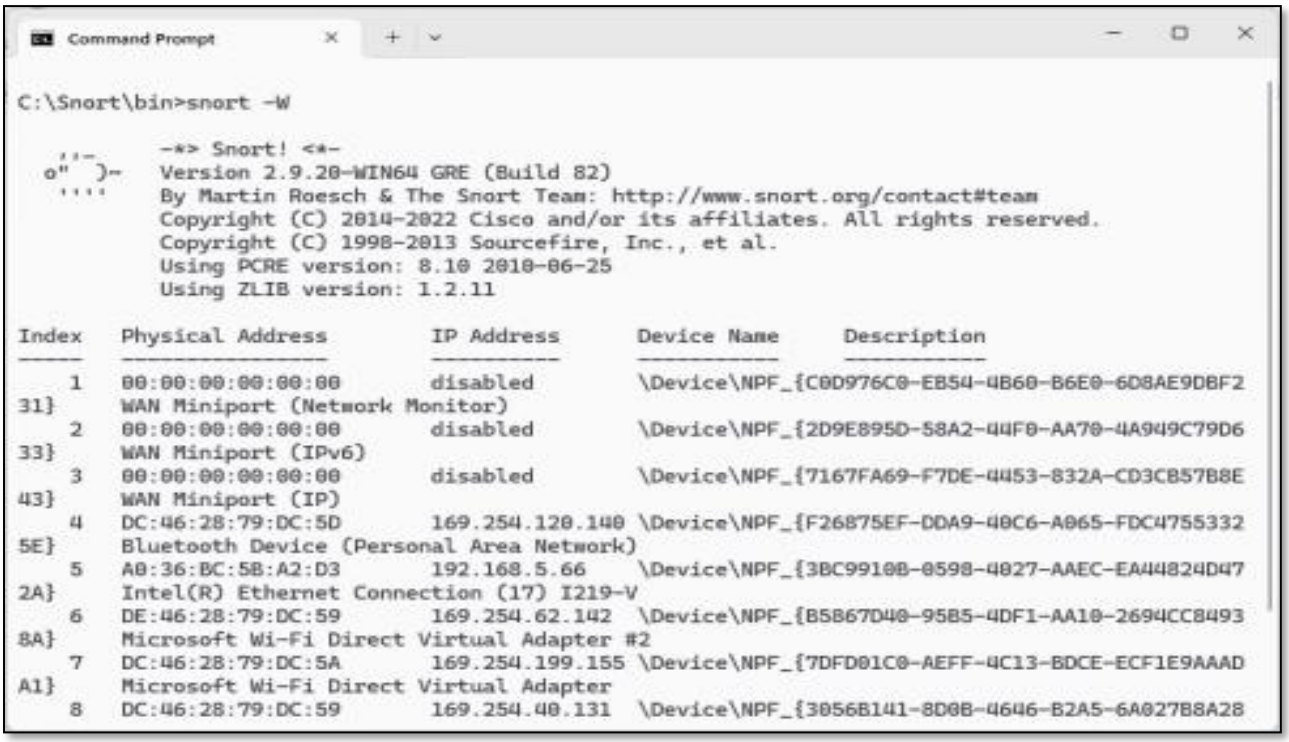


**Observation:**

The output displays Snort's usage information and available command-line options. These options help users configure packet sniffing, logging, alerts, and intrusion detection.

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### 3. Listing Available Network Interfaces (snort -W)

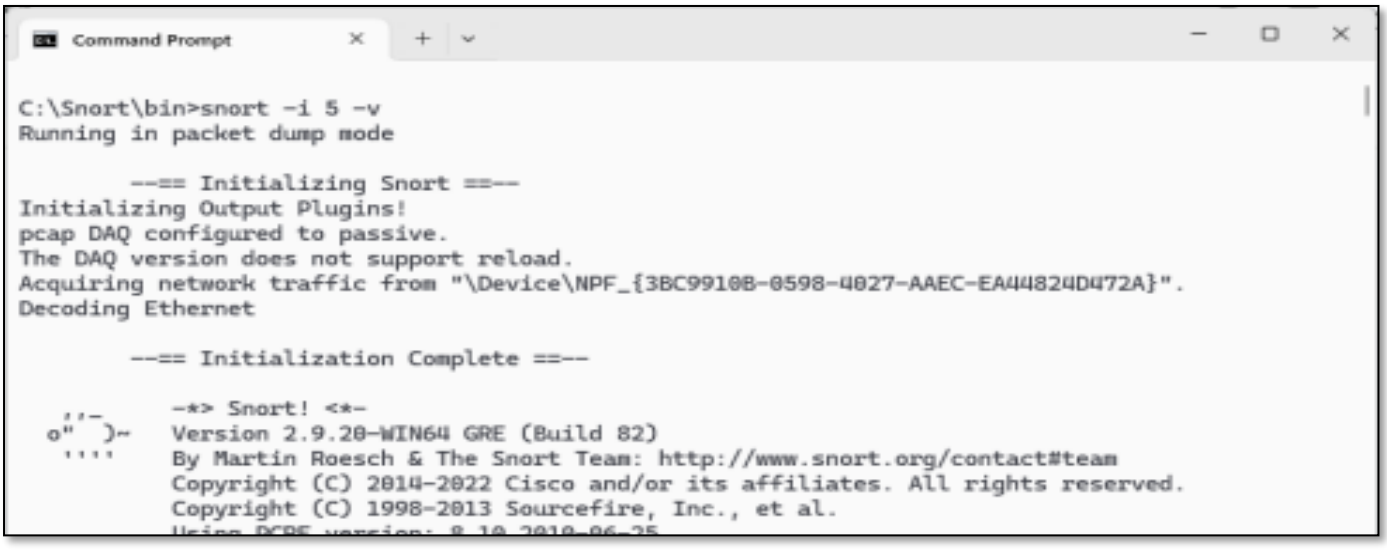



**Observation:**

The command lists the available network interfaces along with their index numbers, physical addresses, IP addresses, device names, and descriptions. The interface index can be used to select a network interface for packet capture.

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### 4. Running Snort in Verbose Mode (snort -i 5 -v)



```
===============================================================================
Packet I/O Totals:
   Received:          239
   Analyzed:          124 ( 51.883%)
    Dropped:            0 (  0.000%)
   Filtered:            0 (  0.000%)
Outstanding:          115 ( 48.117%)
   Injected:            0
===============================================================================
Breakdown by protocol (includes rebuilt packets):
        Eth:          124 (100.000%)
       VLAN:            0 (  0.000%)
        IP4:           56 ( 45.161%)
       Frag:            0 (  0.000%)
```

```
===============================================================================

Memory Statistics for File at:Fri Oct 18 08:55:49 2024

Total buffers allocated:           0
Total buffers freed:               0
Total buffers released:            0
Total file mempool:                0
Total allocated file mempool:      0
Total freed file mempool:          0
Total released file mempool:       0

Heap Statistics of file:
          Total Statistics:
               Memory in use:              0 bytes
                No of allocs:              0
                 No of frees:              0
===============================================================================
Snort exiting

C:\Snort\bin>
```

**Observation:**

Snort initializes on the selected network interface and begins capturing network traffic. The output confirms that initialization is complete and displays information about the selected interface.

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**5. Running Snort in Verbose Mode with Packet Data (snort -i 5 -vd)**

```
    Bad TTL:            0 (  0.000%)
     S5 G 1:            0 (  0.000%)
     S5 G 2:            0 (  0.000%)
      Total:          515
===============================================================================

Memory Statistics for File at:Mon Oct 21 15:16:28 2024

Total buffers allocated:           0
Total buffers freed:               0
Total buffers released:            0
Total file mempool:                0
Total allocated file mempool:      0
Total freed file mempool:          0
Total released file mempool:       0

Heap Statistics of file:
          Total Statistics:
               Memory in use:              0 bytes
                No of allocs:              0
                 No of frees:              0
===============================================================================
Snort exiting

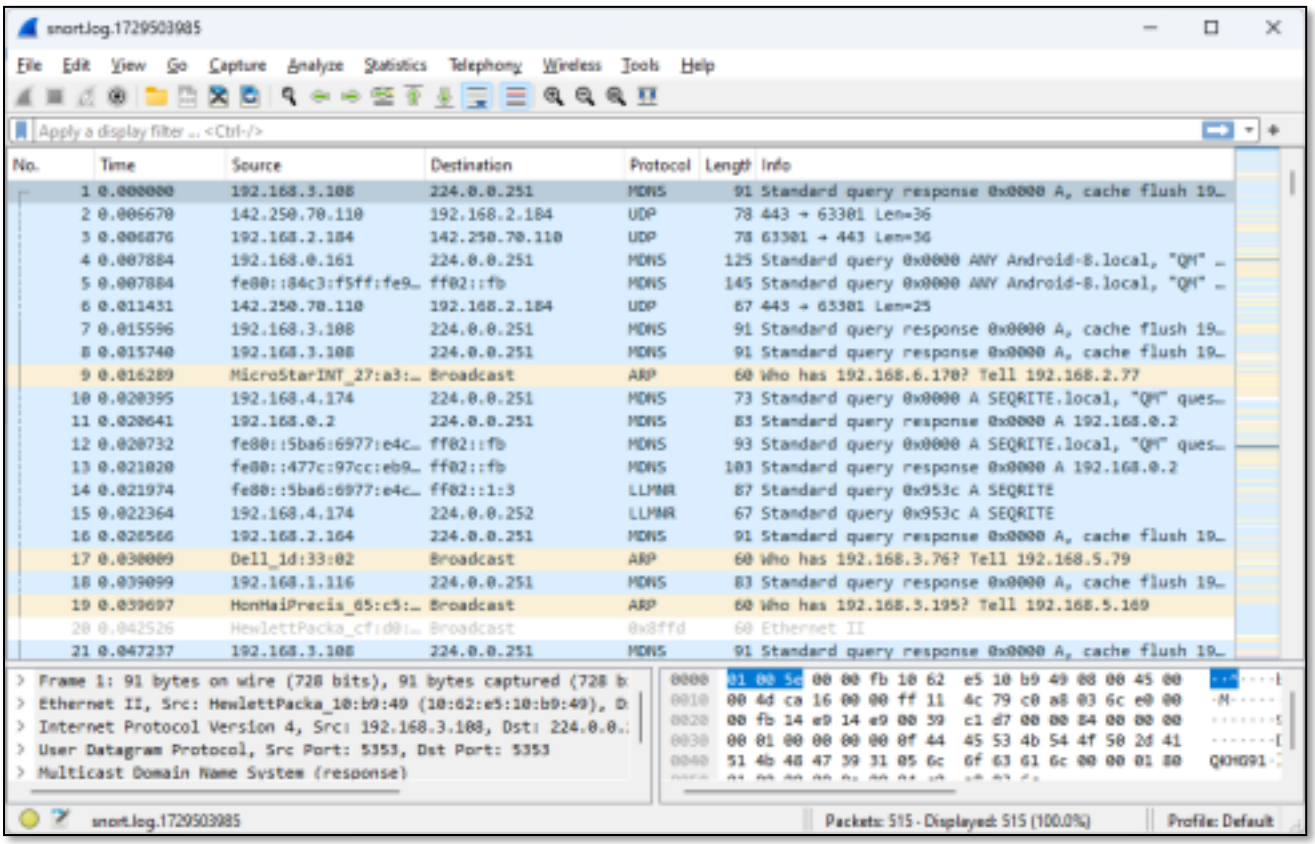
C:\Snort\bin>
```

**Observation:**

The output displays packet input/output statistics, including the number of packets received and analyzed, along with a protocol-wise breakdown. Snort also displays memory statistics before terminating.

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## 6. Viewing the Generated Packet Log in Wireshark

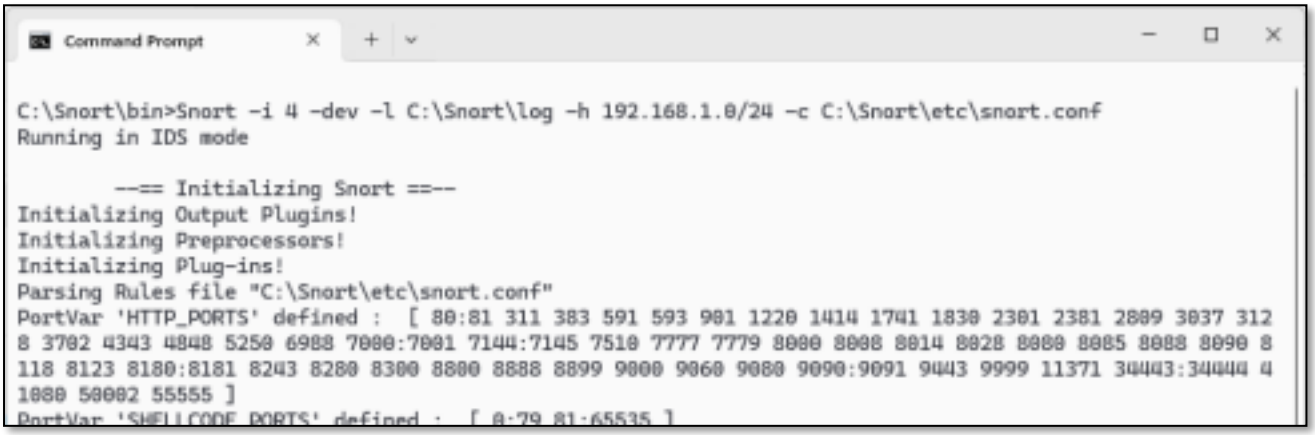



**Observation:**

Wireshark displays the captured network packets from the log file, including their timestamps, source and destination addresses, protocols, and lengths. Selecting a packet reveals its decoded protocol information and hexadecimal data for detailed analysis.

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## 7. Testing Snort IDS Mode Using the Configuration File



```
PortVar 'ORACLE_PORTS' defined :  [ 1024:65535 ]
PortVar 'SSH_PORTS' defined :  [ 22 ]
PortVar 'FTP_PORTS' defined :  [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined :  [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined :  [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381
2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 808
5 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371
34443:34444 41080 50002 55555 ]
PortVar 'GTP_PORTS' defined :  [ 2123 2152 3386 ]
Detection:
   Search-Method = AC-Full-Q
    Split Any/Any group = enabled
    Search-Method-Optimizations = enabled
    Maximum pattern length = 20
ERROR: C:\Snort\etc\snort.conf(247) Could not stat dynamic module path "/usr/local/lib/snort_dynamicpre
processor/": No such file or directory.

Fatal Error, Quitting..
Could not create the registry key.
```

**Observation:**

Snort starts in IDS mode and reads the configuration file, displaying the configured port variables and detection settings. However, execution terminates with a fatal error related to the dynamic preprocessor path and registry key creation, indicating that the IDS configuration did not complete successfully.

---

**8. Inspecting the Snort Configuration File (snort.conf)**

Snort.conf

```
################################################## # This file contains a
sample snort configuration.

# You should take the following steps to create your own custom
configuration:

# 1) Set the network variables.

# 2) Configure the decoder

# 3) Configure the base detection engine

# 4) Configure dynamic loaded libraries

# 5) Configure preprocessors

# 6) Configure output plugins

# 7) Customize your rule set

# 8) Customize preprocessor and decoder rule set # 9)
Customize shared object rule set

# 9) Customize shared object rule set

##################################################

################################################## # Step #1:
```

```
Set the network variables. For more information, see README.variables

##################################################

# Setup the network addresses you are protecting
ipvar HOME_NET any

# Set up the external network addresses. Leave as "any" in most
situations

ipvar EXTERNAL_NET any

# List of DNS servers on your network

ipvar DNS_SERVERS $HOME_NET

# List of SMTP servers on your network

ipvar SMTP_SERVERS $HOME_NET

# List of web servers on your network

ipvar HTTP_SERVERS $HOME_NET
```

**Observation:**

The snort.conf file contains the settings required to configure Snort for a specific network environment. It describes the main configuration steps and defines network variables such as HOME_NET, EXTERNAL_NET, DNS_SERVERS, SMTP_SERVERS, and HTTP_SERVERS, which help specify the protected network and relevant servers.